

The Constants of Nature

$\epsilon_{\rm h}:\text{K}=\text{kelvin}\frac{\left(\frac{T_{\rm p}}{m_{\rm p}}\right)}{m_{\rm p}}\left(1-14C_{\rm U}^{-4}\mathscr{K}_{\Theta}\right)\boxtimes$	$G=\left(\frac{l_{\rm p}^3}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\frac{\left(4\pi\right)}{18}\right)\mathscr{K}_{\circ}^3\boxtimes$	$\epsilon_{\rm p}=\frac{\text{joule}}{\text{GeV}}\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2}\right)\left(1-\frac{1}{2}Re\left(l^{i'v}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm h}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}q_{\rm p}}{l_{\rm p}m_{\rm e}}\right)\left(1+\sqrt{\frac{4}{D_{\rm Do}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$c=\left(\frac{l_{\rm p}}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm Gl}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$F_{90}=\text{farad}\left(1+\frac{C_{\rm d}}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$K_j=\frac{\mathscr{K}_1}{\pi}\left(\frac{t_{\rm p}q_{\rm p}}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$g_{\rm de}=\frac{2\pi}{\text{C}}\left(1+\frac{16}{l_{\rm p}^2}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\boxtimes\right)$
$\text{K}:\epsilon_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_1^4}\left(\frac{t_{\rm p}}{l_{\rm p}m_{\rm e}}\right)\left(1+14C_{\rm U}^{-4}\mathscr{K}_{\Theta}\right)\boxtimes$	$\frac{G}{hc}=\text{GeV}^2\left(\frac{t_{\rm p}^4}{l_{\rm p}^3m_{\rm p}^2}\right)\left(1-\frac{\left(4\pi\right)}{18}\right)\mathscr{K}_{\circ}^3\boxtimes$	$R_{\rm K}=\frac{2\pi}{\mathscr{K}_1^2}\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(l^{i'v}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}q_{\rm p}}{l_{\rm p}m_{\rm e}}\right)\left(1+\sqrt{\frac{4}{D_{\rm Do}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm Gl}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$H_{90}=\text{henry}\left(1-\frac{C_{\rm d}}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$e^-=\mathscr{K}_1\left(\frac{t_{\rm p}q_{\rm p}}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$\frac{\mu_{\rm de}}{8K}=\frac{2\pi}{8K}\left(1+\frac{16}{l_{\rm p}^2}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\boxtimes\right)$
$\Delta_{\rm vcs}=\frac{E_1}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\mathscr{K}_{\Theta}^2\boxtimes\right)$	$m_{\rm e}=2V_{\rm fe}\left(\frac{m_{\rm p}^6}{l_{\rm p}^2}\right)\left(1+\sqrt{\frac{18}{2V_{\rm fe}}}\mathscr{K}_{\circ}^3\boxtimes\right)$	$Z_0=4\pi\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(l^{i'v}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathscr{K}_1^2}\left(\frac{l_{\rm p}m_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$\Omega_{90}=\text{ohm}\left(1-\frac{C_{\rm d}}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2m_{\rm p}}{l_{\rm p}^2q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8K}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{l_{\rm p}^2}\left(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2\right)\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\mathscr{K}_{\Theta}^2\boxtimes\right)$	$m_{\rm a}=\frac{\omega_{\rm z}^2}{\mathscr{K}_1^3}\sqrt{\frac{16}{2V_{\rm fe}}}\left(\frac{m_{\rm p}^6}{\text{kilogram}^3}\right)\left(1+\frac{\left(75\right)}{15}\mathscr{K}_{\circ}^3\boxtimes\right)$	$Z_{\rm p}=\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(l^{i'v}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\sigma_{\rm p}=\sqrt{\frac{32}{18}}2\pi\mathscr{K}_1^4\left(\frac{t_{\rm p}m_{\rm p}}{l_{\rm p}^2}\right)^2\left(1-2\sqrt{6C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\text{A}_{\rm mass}}{l_{\rm p}^2T_{\rm p}m_{\rm p}}\right)^{3/2}\frac{K E_{\rm p}}{T_{\rm p}P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$b'=\frac{1}{2\pi}\left(\frac{v_{\rm peak}}{t_{\rm p}T_{\rm p}}\right)\left(1+\frac{1}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2\right)\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\mathscr{K}_{\Theta}^2\boxtimes\right)$	$m_{\rm m}=\frac{\mathscr{K}_1^{-5}}{18}\sqrt{\frac{4s}{2\pi}}\left(\frac{m_{\rm p}^4}{\text{kilogram}^3}\right)\left(1+3\omega_1^3\mathscr{K}_{\circ}^3\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\rm p}}{l_{\rm p}}\right)\left(1+\frac{1}{2}Im\left(l^{i'v}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm s}^{'}}{S}=\frac{2\mathscr{K}_1\mathscr{K}_2^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{G_{\rm f}}{(hc)^3}=\sqrt{\frac{32}{18}}\frac{\left(2\pi\right)^2}{\text{kilogram}}\left[\frac{\text{joule}}{\text{GeV}}\right]^2\left(\frac{t_{\rm p}^4}{l_{\rm p}^2m_{\rm p}}\right)\left(1+28\sqrt{6C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\text{A}_{\rm mass}}{l_{\rm p}^2T_{\rm p}m_{\rm p}}\right)^{3/2}\frac{K E_{\rm p}}{T_{\rm p}P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\boxtimes\right)$	$\frac{h}{e}=2\pi\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{l_{\rm p}^2m_{\rm p}}{l_{\rm p}^2q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\right)\boxtimes$	$\mathscr{K}_{\rm B}=\frac{1}{2\pi}\left(\frac{1}{t_{\rm p}T_{\rm p}}\right)\left(1+\frac{1}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2\right)\boxtimes\right)$
$\text{Hz}:\text{J}=\frac{2\pi}{\text{second}}\left(\frac{l_{\rm p}^2m_{\rm p}}{t_{\rm p}}\right)\left(1+\sqrt{\zeta(2)}\mathscr{K}_{\Theta}^2\boxtimes\right)$	$K_{\rm cd}=\frac{\mathscr{K}_1^{-7}}{f_1}\sqrt{\frac{4s}{2\pi}}\left(\frac{t_{\rm p}^2}{l_{\rm p}^2A_{\rm mass}}\right)\left(1+3\omega_2^3\mathscr{K}_{\circ}^3\boxtimes\right)$	$A_{\rm vel}=\mathscr{K}_1^2\left(\frac{t_{\rm p}}{l_{\rm p}}\right)\left(1-\frac{1}{2}Im\left(l^{i'v}\right)\mathscr{K}_3$					

$t_p = 5.3912583683213 \dots \times 10^{-46}$ s	Planck time
$l_p = 1.61625918175645 \dots \times 10^{-38}$ m	Planck length
$q_p = 1.87554596713962 \dots \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \dots \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \dots \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \dots \times 10^{-7}$	hyperbolic inversion boundary
5	number of dimensions
7	break in scale symmetry
$\pi_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\pi_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\pi_3 = -1.8764943900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\pi_4 = -1.8764943900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\pi_5 = 4.47826244916751 \dots$	hyperbolic radius constant
$\pi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{00} = 1.01494160640965 \dots$	Gieseking's constant
$V_6 = 0.29598321821930 \dots$	figure eight knot hyperbolic volume
$K = 0.01596559471727190 \dots$	Catalan's constant
$C_d = 0.2916506090430818 \dots$	domino tiling constant
$C_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649747018528596 \dots + 1.3249706214087 \dots i$	omega_1 constant
$\omega_2 = 1.529594037057192 \dots$	omega_2 constant
$i^{i^{i^i}} = 0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\phi_1 = (-1)^{1/3}$	imaginary golden ratio
$G_9 = 1.15872847301812 \dots$	tether length for grazing half the unit circle
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$S = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.244115110858423 \dots$	arc length of the unit lemniscate
$L = 2.620257554292119 \dots$	lemniscate constant
$L_1 = 1.31102877146205 \dots$	1 st lemniscate constant
$L_2 = 0.5949701117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.889213084793979 \dots$	ubiquitous constant
$G_{06} = 0.834626841674073 \dots$	Gauss's constant
$C_{CF} = 0.697777645664095 \dots$	continued fraction constant
$R_{FR} = 0.9981360445980509 \dots$	Ramanujan's 1 st continued fraction constant
$F_1 = 1.22674201072035 \dots$	Fibonacci factorial constant
$M = -1.74756459463318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822862549678847 \dots$	Sierpinski's constant
$C_{JP} = 2.40488256540057572 \dots$	1 st root of the Bessel function
$J_{001} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{N0} = 0.739085133216160 \dots$	Dottie number
$L = 0.662743419349181 \dots$	Laplace limit
$\alpha_F = 2.50290787590589 \dots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.246497960371714 \dots$	silver constant
$P = 1.32471795724474 \dots$	plastic constant
$Q_0 = 0.605443657196732 \dots$	QRS constant
$C_{QA} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{\text{ISM}} = 0.353236371854995 \dots$	Hafner-Sarnak-McCurley constant
$T_0 = 2.731500000000000 \dots \times 10^2$ kelvin	freezing point of water
$P_0 = 1.000000000000000 \dots \times 10^5$ pascal	1 bar
$P_1 = 1.01325003754773 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules	energy associated with $\Delta\epsilon_{\text{CS}}$
$f_1 = 5.40000000000000 \dots \times 10^{14}$ Hz	frequency of max/light wavelength
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.8214393712207 \dots$	peak emission of spectral flux/unit frequency
$\log(x) =$ natural logarithm	
$\Gamma(x) =$ gamma function	ln = lumen = cd · sr
$\zeta(x) =$ Riemann zeta function	$N =$ newton = m kg/s ²
$\ln =$ derangement function	$N =$ noether = m kg/s
	$\nu =$ ohm = m ² kg/s ² C ²
units	cosmetic units
$m =$ meter	$A =$ ampere = C/s
$s =$ second	$F =$ farad = s ² C ² /m ² kg
$C =$ coulomb	$H =$ henry = m ² kg/C ²
$K =$ kelvin	$\text{Hz} =$ hertz = 1/s
$\text{kg} =$ kilogram	$J =$ joule = m ² kg/s ²
derived units	
MHz = megahertz = 1.0000000000000000... × 10 ⁶ 1/s	
fm = femtometer = 1.0000000000000000... × 10 ⁻¹⁵ m	
eV = electron-volt = 1.0000000000000000... eV	
MeV = megaelectron-volt = 1.0000000000000000... × 10 ⁶ eV	
GeV = gigaelectron-volt = 1.0000000000000000... × 10 ⁹ eV	
$\frac{r_n}{r_e} = \left(\frac{4\pi}{rS}\right)^2$	$\frac{r_n}{r_e} = \sqrt{\frac{C_0}{4}}$
$\frac{r_n}{r_e} = \sqrt{\frac{C_0}{4}}$	$\frac{r_e}{q_0} = \sqrt{\frac{2\pi}{\log(36)}}$
$\epsilon_p = \frac{l^2 m_p}{\epsilon_r r^2}$	
external transform space	Physics Monastery
$V_{2\pi} = \left\{ \left(\frac{4\pi}{5i} \right) \left(\frac{4\pi}{15} \right) \left(\frac{4\pi}{35} \right) \left(\frac{4\pi}{49} \right) \left(\frac{4\pi}{32} \right) \left(\frac{4\pi}{8} \right) \right\}^4$	Thad Roberts
figure eight knot hyperbolic volume	January-17-2026